

Velum.

Holder-attested compliance for Confidential Tokens on Stellar.

A regulated fund can hide its register of holders — and still prove it obeys the rules. Without revealing the amount to anyone. **Including the verifier.**

Tokenizing a fund makes it **less** private than the paperwork it replaces.

In the filing cabinet

The register of holders sits with the administrator. Who subscribed, how much, when they left — none of it is public.

On a public ledger

All of it is. Ten institutional positions de-anonymize by correlating subscription timing with off-chain events — and the information has a paying audience.

Confidential Tokens fix the balances. They do not fix compliance.

Compliance has two halves. Upstream ships the primitives for both, wired to neither.

THE QUESTION

UPSTREAM TODAY

WHAT A REGULATED FUND NEEDS

Who may hold?

A flat **address allowlist**. No reason, no expiry, no attesting party.

A **valid claim** — qualified investor — issued by an approved certifier. Their issue #766. Still open.

How much may they hold?

Structurally impossible. `is_authorized(account, token) -> bool` sees no value. Balances are commitments.

Minimum ticket. Concentration ceiling. Per-investor cap. Every rule that mentions a number.

So the check has to move.

Under **FHE**, a contract compares ciphertexts with no witness — the check fits inside the transfer. Under **Pedersen commitments**, a proof needs a witness. And for any balance, exactly one party holds it.

A sender cannot prove anything about the recipient's balance. It has no opening for it — and binding guarantees it cannot find another.

Quantitative compliance cannot be **transaction-gated**. It must be **holder-attested** — the party that can prove is the party whose balance it is.

Four pieces.

COMPONENT	WHAT IT DOES
velum-policy	The token's gate asks an ERC-3643/T-REX identity registry instead of a list. One KYC registry governs the public asset and the confidential wrapper.
disclose_balance_ge	The predicate circuit upstream specifies and does not ship . The holder proves " <i>my position</i> $\geq T$ ". The proof's validity <i>is</i> the assertion — there is nothing to decrypt.
velum-attest	On-chain verification — which the same spec marks out of scope. Every public input is read from the token and the profile; the prover supplies only the proof.
circuits/seize	Answers their open question on individual clawback : premise verified, circuit built, partial seizure demonstrated.

Three things you can watch happen.

01 The explorer goes blind

Addresses visible, amounts encrypted.
The register stops being public.

02 A wallet without a claim is refused

Not because it is on a list — because it carries no qualified-investor attestation.

03 A holder proves the threshold

A contract verifies it on-chain. Nobody learns the amount — the verifier included.

And a false claim? **It cannot even be constructed.** The witness fails before a proof exists.

MEASURED ON OUR HARDWARE · NOT ESTIMATED

The numbers.

0.25 s to prove a position against a regulatory threshold

14 592 B proof size · 6 public inputs

93 opcodes the seize circuit — cheaper than the transfer circuit itself

+3 % cost of adding a quantitative rule inside the transfer circuit

66 tests across four circuit packages, all passing

Because they wrote it down.

COMPLIANCE.md:218 — OpenZeppelin, on why individual clawback is unsolved:

"Because the clawback circuit does not have access to `vk_A` ... The follow-up revision will pin down whether `r_s` is supplied as a private witness with an auxiliary opening proof or **derived in-circuit from a separately escrowed value**."

They name the obstacle **and** list the remedy. We take the second option and say what to escrow: `vk_A` itself — then verify it in code.

Every claim we make about their product is sourced to a file and a line in Appendix A. Trait signatures and verbatim quotes — not our characterisation.

The honest part.

Not production

Developer-preview primitives; the UltraHonk verifier is unaudited. Testnet only, no real value.

Not finished

The clawback migration — a register-circuit change and re-registration — is priced, not built.

Not solved

Predicates over **encrypted aggregates across accounts** — concentration by tranche, subordination ratio between senior and subordinated — remain open. As far as we can determine, on any chain.

Not invented

We implement their specification. The novelty claim is bounded: no public implementation exists in any repository, branch, tag or history.

WHERE THIS GOES

"Regulated" and "public" stop being opposites.

The anchor case is a Brazilian receivables fund under CVM 175 — senior and subordinated classes, a register that is public today. The kit serves any regulated issuer with the same paradox.

MIT · open source

github.com/ThaisFReis/velum

whitepaper · circuits · contracts

five upstream findings